1. $(x^a)' =$

2. $(a^x)' =$

3. $(e^x)' =$

4. $(\log_a x)' =$

5. $(\ln|x|)' =$

6. $(\sin x)' =$

7. $(\cos x)' =$

8. $(\tan x)' =$

9. $(\cot x)' =$

10. $(\sec x)' =$

11. $(\csc x)' =$

12. $(\ln|\cos x|)' =$

13. $(\ln|\sin x|)' =$

14. $(\ln|\sec x + \tan x|)' =$

15. $(\ln|\csc x - \cot x|)' =$

16. $(\ln|\csc x + \cot x|)' =$

17. $(\arcsin x)' =$

18. $(\arccos x)' =$

19. $(\arctan x)' =$

20. $(\operatorname{arccot} x)' =$

21. $\left[\ln(x + \sqrt{x^2+1})\right]' =$

22. $\left[\ln(x + \sqrt{x^2-1})\right]' =$

23. $\left[\ln\left(x + \sqrt{x^2+a^2}\right)\right]' =$

24. $(\frac{1}{2}\arcsin x + \frac{x}{2}\sqrt{1-x^2})' =$

25. $\left[\ln\left(x + \sqrt{x^2-a^2}\right)\right]' =$

26. $(x\ln x)' =$

27. $(\frac{1}{x}\ln x)' =$

28. $\left(e^{ax+b}\right)^{(n)} =$

29. $\left[\sin(ax+b)\right]^{(n)} =$

30. $\left[\cos(ax+b)\right]^{(n)} =$

31. $\left[\ln(ax+b)\right]^{(n)} =$

32. $\left(\frac{1}{ax+b}\right)^{(n)} =$

33. $(\frac{1}{1+x})^{(n)} =$

34. $\int x^k \,\mathrm{d}x =$

35. $\int \frac{1}{x} \,\mathrm{d}x =$

36. $\int e^x \,\mathrm{d}x =$

37. $\int a^x \,\mathrm{d}x =$

38. $\int \sin x \,\mathrm{d}x =$

39. $\int \cos x \,\mathrm{d}x =$

40. $\int \tan x \,\mathrm{d}x =$

41. $\int \cot x \,\mathrm{d}x =$

42. $\int \frac{1}{\cos x} \,\mathrm{d}x =$

43. $\int \frac{1}{\sin x} \,\mathrm{d}x =$

44. $\int \frac{1}{\cos^2 x} \,\mathrm{d}x =$

45. $\int \frac{1}{\sin^2 x} \,\mathrm{d}x =$

46. $\int \sec x \tan x \,\mathrm{d}x =$

47. $\int \csc x \cot x \,\mathrm{d}x =$

48. $\int \frac{1}{1+x^2} \,\mathrm{d}x =$

49. $\int \frac{1}{a^2+x^2} \,\mathrm{d}x =$

50. $\int \frac{1}{\sqrt{1-x^2}} \,\mathrm{d}x =$

51. $\int \frac{1}{\sqrt{a^2-x^2}} \,\mathrm{d}x =$

52. $\int \sqrt{1-x^2} \,\mathrm{d}x =$

53. $\int \sqrt{a^2-x^2} \,\mathrm{d}x =$

54. $\int \frac{1}{\sqrt{x^2+1}} \,\mathrm{d}x =$

55. $\int \frac{1}{\sqrt{x^2+a^2}} \,\mathrm{d}x =$

56. $\int \frac{1}{\sqrt{x^2-1}} \,\mathrm{d}x =$

57. $\int \frac{1}{\sqrt{x^2-a^2}} \,\mathrm{d}x =$

58. $\int \sqrt{1+x^2} \,\mathrm{d}x =$

59. $\int \frac{1}{x^2 - a^2} \, \mathrm{d}x =$

60. $\int \frac{1}{a^2 - x^2} \, \mathrm{d}x =$

61. $\int \sin^2 x \, \mathrm{d}x =$

62. $\int \cos^2 x \, \mathrm{d}x =$

63. $\int \tan^2 x \, \mathrm{d}x =$

64. $\int \cot^2 x \, \mathrm{d}x =$

65. $\int_0^{\frac{\pi}{2}} \sin^n x \, \mathrm{d}x =$

66. $\int_0^{\frac{\pi}{2}} \cos^n x \, \mathrm{d}x =$

67. $\int_0^{\pi} \sin^n x \, \mathrm{d}x =$

68. $\int_0^{\pi} \cos^n x \, \mathrm{d}x =$

69. $\int_0^{2\pi} \sin^n x \, \mathrm{d}x =$

70. $\int_0^{2\pi} \cos^n x \, \mathrm{d}x =$

71. $\int e^{ax} \cos bx \, \mathrm{d}x =$

72. $\int e^{ax} \sin bx \, \mathrm{d}x =$

1. $\sin(\frac{\pi}{2} \pm \alpha) =$

2. $\sin(\pi \pm \alpha) =$

3. $\cos(\frac{\pi}{2} \pm \alpha) =$

4. $\cos(\pi \pm \alpha) =$

5. $\sin(\alpha \pm \beta) =$

6. $\cos(\alpha \pm \beta) =$

7. $\tan(\alpha \pm \beta) =$

8. $\tan(\frac{\pi}{4} - \alpha) =$

9. $\cot(\alpha \pm \beta) =$

10. $\sin 2\alpha =$

11. $\cos 2\alpha =$

12. $\sin 3\alpha =$

13. $\cos 3\alpha =$

14. $\sin^2 \frac{\alpha}{2} =$

15. $\cos^2 \frac{\alpha}{2} =$

1. $e^x =$

2. $\sin x =$

3. $\cos x =$

4. $\arctan x =$

5. $\frac{1}{1-x} =$

6. $\frac{1}{1+x} =$

7. $\ln(1+x) =$

8. $(1+x)^a =$

9. $\tan x =$

$\arcsin x =$

1. 七种未定式：

2. 等差数列通项公式：
3. 等差数列前 n 项和：
4. 等比数列通项公式：
5. 等比数列前 n 项和：
6. $2f(x) \cdot f'(x) =$
7. $[f'(x)]^2 + f(x)f''(x) =$
8. $[f'(x) + f(x)\varphi'(x)]e^{\varphi(x)} =$
9. $f'(x) + f(x) =$
10. $f'(x) - f(x) =$
11. $f'(x) + kf(x) =$
12. 曲率公式：
13. 曲率半径公式：
14. 牛顿-莱布尼茨公式：
15. 区间再现公式：
16. $u''v + 2u'v' + uv'' =$
17. $\dfrac{xf'(x) - f(x)}{x^2} =$
18. $\dfrac{f''(x)f(x) - [f'(x)]^2}{f^2(x)} =$
19. $\dfrac{f''(x)f(x) - [f'(x)]^2}{f^2(x)} =$
20. $\dfrac{f'(x)}{f(x)} =$

1. 有界与最值定理：
2. 介值定理：
3. 平均值定理：
4. 零点定理：
5. 费马定理：
6. 导数零点定理：
7. 罗尔定理：
8. 拉格朗日中值定理：
9. 柯西中值定理：

1. 若 $a_1, a_2, \cdots, a_m \geq 0$，则 $\lim\limits_{n\to\infty} \sqrt[n]{a_1^n + a_2^n + \cdots + a_m^n} =$

2. 三角不等式

3. 反三角不等式

4. 基本不等式

5. 三元均值不等式

6. $0 < a < x < b, 0 < c < y < d$

7. $e^x \quad 1 + x$

8. $\ln x \quad x - 1\ (x > 0)$

9. $\dfrac{1}{1+x} \quad \ln\left(1 + \dfrac{1}{x}\right) \quad \dfrac{1}{x}, (x > 0)$

10. $\dfrac{x}{1+x} \quad \ln(1+x) \quad x, (x > 0)$

11. $0 < x < \dfrac{\pi}{2}$

12. $\dfrac{1}{n^2+n} \quad \dfrac{1}{n^2+k} \quad \dfrac{1}{n^2+1}$

13. $n \quad \sqrt{n^2+k} \quad \sqrt{n^2+n}$

14. $x > 0, \sin x \quad x$

15. $0 < x < \dfrac{\pi}{4}$

16. $x > 0, \arctan x \quad x \quad \arcsin x$